

CSE331 Assignment 1 Part B

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- There will be two parts in this assignment. Part A and Part B. Both need to be submitted together online in pdf format in a google form that will be published later.

- Deadline for entire Assignment 1 submission:

13 November, 2025, Thursday, 11:59 pm.

- Only one submission per group.
- The submitted document should have clear and legible scans.

1. (15 marks) Write the Regular Expressions for the following languages:

- a. Every 1 is followed by odd number of 0's
- b. Doesn't begin with 110
- c. Contains substring 110 but not 00
- d. Contains 1 in every 4th position
- e. Doesn't contain exactly 3 0's
- f. Has the symbol it starts with in every even position

2. (20 marks) Convert the Regular Expressions to NFA:

- a. $(0+1)^*010(0+1)^*$
- b. $1^*(01^+)^*01^*$
- c. $(0+01^*0)^*01^*0$
- d. Groups with 3 members need to convert the Regular Expression obtained in 1(b), 1(c), 1(d) and 1(f) into NFA

3. (15 marks) Convert the following DFA into Regular Expression using the State Elimination Method. You must follow the sequence q_2, q_4, q_1, q_3 . You need to show detailed steps of the process.

